

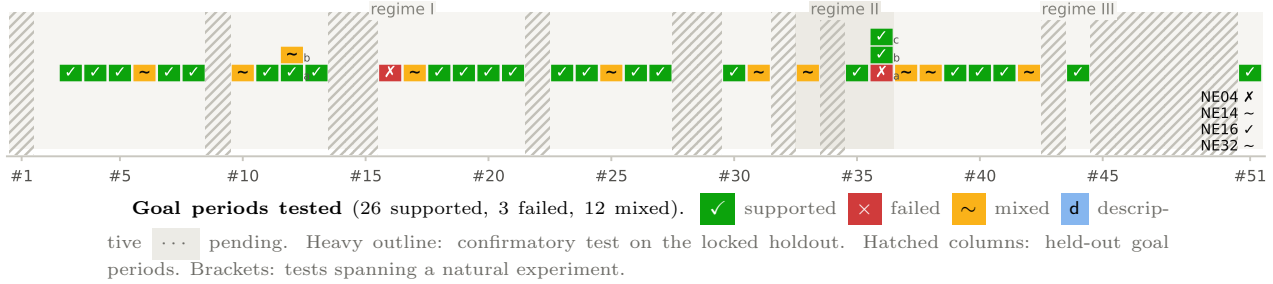
H71: Memory is a first-order homeostat

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. Memory size is regulated toward an agent set point on the consolidation clock, without overshoot; the loop is a fast near-deadbeat compression around a slowly drifting target, not a first-order homeostat.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH269 claims that memory size is a first-order homeostat with overshoot after erasures and a gain change at NE41. H71 fits the post-compression memory size from `memory_stats` (62k non-holdout cycles, 37 units, regimes I–III). Each regime-III consolidation writes an append snapshot, then a compression snapshot. **Memory is a homeostat, but not first order, and it does not overshoot.** Size reverts toward an agent set point in 37/37 units (ϕ^+ pooled 0.67 [0.63, 0.71] per cycle). H09's overshoot is the mixed sampling: the mixed series gives $\phi < 0$ in 10/10 regime-III units while $\phi^+ > 0$. Forced and voluntary consolidations give the same ϕ^+ . AR(2) terms are positive (median +0.19). NE14 raises the set point (+17–31%). The holdout script is not run.

PHYSICAL MODEL

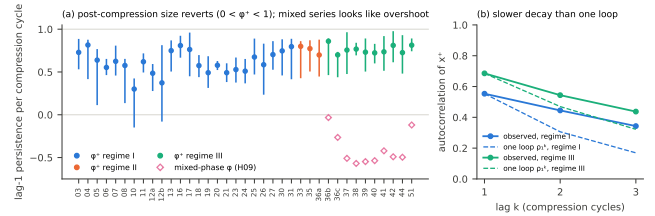
Model 04 (a maintained store) with model 05's import and removal bookkeeping. The degree of freedom is x_n , the log size of an agent's memory after compression n . Growth adds Δ_n (appends), and compression maps the peak y_n back toward a set point μ_i . A single first-order loop gives an AR(1) with gain $1 - \phi$, where $\phi = b(1 + c)$. The rivals are a random walk ($\phi = 1$), a deadbeat template ($\phi = 0$) and an over-correcting controller ($\phi < 0$). A post hoc model adds a slowly drifting set point m_n .

$$y_n = x_{n-1} + \Delta_n, \quad \Delta_n = d_i + c(x_{n-1} - \mu_i) + \xi_n$$

$$x_n = a_i + b y_n + \zeta_n, \quad \phi = b(1 + c)$$

$$x_n - \mu_i = \phi(x_{n-1} - \mu_i) + \varepsilon_n$$

post hoc: $x_n = m_n + e_n, \quad m_n = \rho_s m_{n-1} + \eta_n$



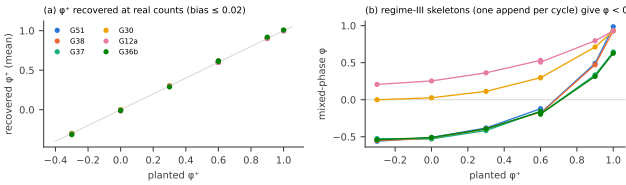
Observables. (a) ϕ^+ per unit (agent bootstrap, half-panel jackknife): all between 0 and 1. Open diamonds give H09's mixed-phase statistic for regime III. It is negative in every unit, which looks like overshoot. (b) Autocorrelation of x^+ at lags 1–3, averaged over units, against the one-loop prediction ρ_1^k (dashed). The slower decay shows a second, slow timescale.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a scaffold builder, memory size is a regulated variable with a per-agent set point (agent identity holds 91% of its variance, the lab 41%). Agents restore it within a few consolidations, so a memory-size alarm needs a per-agent baseline, not a global cap. Forced erasures do not destabilize it. The NE14 scaffold change raised the set point by about 20% and left the per-cycle gain almost unchanged, so a prompt change moves the target, not the loop. A monitor reading raw `memory_stats` will see a false overshoot ($\phi \approx -0.5$) from the two-snapshot write. Filter to compression rows first. The result is a design fact about the scaffold's memory loop. It does not say what the memory is worth (H15: about 0 at day scale).

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P1 $0 < \phi^+ < 0.9$, CI excl. 0, 1	35/37; pooled 0.67 [0.63, 0.71]; overshoot 0/37	supported
P2 AR(1) beats RW $\geq 70\%$	69% of 353 agent-periods; beats deadbeat 88%	narrowly failed
P3 mixed $\phi < 0$, $\phi^+ > 0$ (III)	10/10; one-loop synthetic matches 0/10	mixed
P4 no overshoot after forced	ϕ_F^+ 0.49–0.79; $ F - V < 0.15$ in 9/10	supported
P5 $ AR(2) < 0.1$	5/37; median +0.19	failed
P6 event clock	29/37 $ slope \leq 0.1$	supported
P7 agent-level gain	across-period r 0.52 (null p95 0.33)	supported
P8 $\phi^+ \approx b(1 + c)$	$b(1 + c)$ median 0.13 vs ϕ^+ 0.67	failed
NE14 / NE04	set point +17–31%, $\Delta\phi^+ +0.12$ / $\Delta\phi^+ +0.15$	mixed / failed
NE16 / NE32	no change / relaxation ≈ 0.9 per cycle	supported / mixed



Second figure. Synthetic one-loop dynamics on six real cycle skeletons. (a) The estimator recovers the planted ϕ^+ (bias ≤ 0.02 ; power 1.0 for an overshoot of -0.3). (b) With one append per cycle (regime III), the mixed-phase ϕ is negative for any planted $\phi^+ \leq 0.6$. This is H09's artifact. A post hoc two-timescale sawtooth matches the observed size in 7/8 units.

GOAL PERIODS IN DETAIL

G03 ✓ supported
G04 ✓ supported
G05 ✓ supported
G06 ~ mixed
G07 ✓ supported
G08 ✓ supported
G10 ~ mixed
G11 ✓ supported
G12a ✓ supported
G12b ~ mixed
+31 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	2	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The two-timescale model is post hoc: it explains P5 and P8 after they failed. Size is a capacity, not content; hashed-line turnover is not analysed. Regime-I cycles are session-scale and differ in structure from regime-III cycles, so pooling across regimes compares unlike cycles. NE04 has 7 agents and a goal change inside the window. NE16 has two days per side. The out-of-sample margin over a random walk is small (median MSE ratio 0.89) because the target drifts. The holdout is not used; #43, the memory goal, is the main confirmation target.

ROUND 2

- **H71-R1** Fit the two-timescale model directly (state-space Kalman fit per agent) and test the set point's drift against task covariates: does the target follow the agent's active repo count or its role (DQ4, DQ6)?
- **H71-R2** Set point vs viability: do agents whose memory is above its set point at a forced erasure recover output faster (H44 re-acquisition, H15 dip)?
- **H71-R3** Run `confirm.py` (#43 is the one goal that acts on the store) after disclosures.
- **H71-R4** Content turnover: the hashed-line Jaccard per cycle as a second order parameter (what is kept, not how much).